Laguna State Polytechnic University

ITEP 414 - System Administration and Maintenance

Installation Guide

Enterprise Server Deployment and Operating System Installation

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Program: BSIT

Section: BSIT-4E

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1. Project Overview

This guide documents the Week 3 enterprise server deployment project. Ubuntu Server was deployed in Oracle VirtualBox for the ABC Startup Solutions scenario, verified after installation, and documented with evidence. The project also includes a BIOS-versus-UEFI study, Ubuntu boot-process flowchart, Windows Server Evaluation installation, and operating-system comparison.

2. Objectives

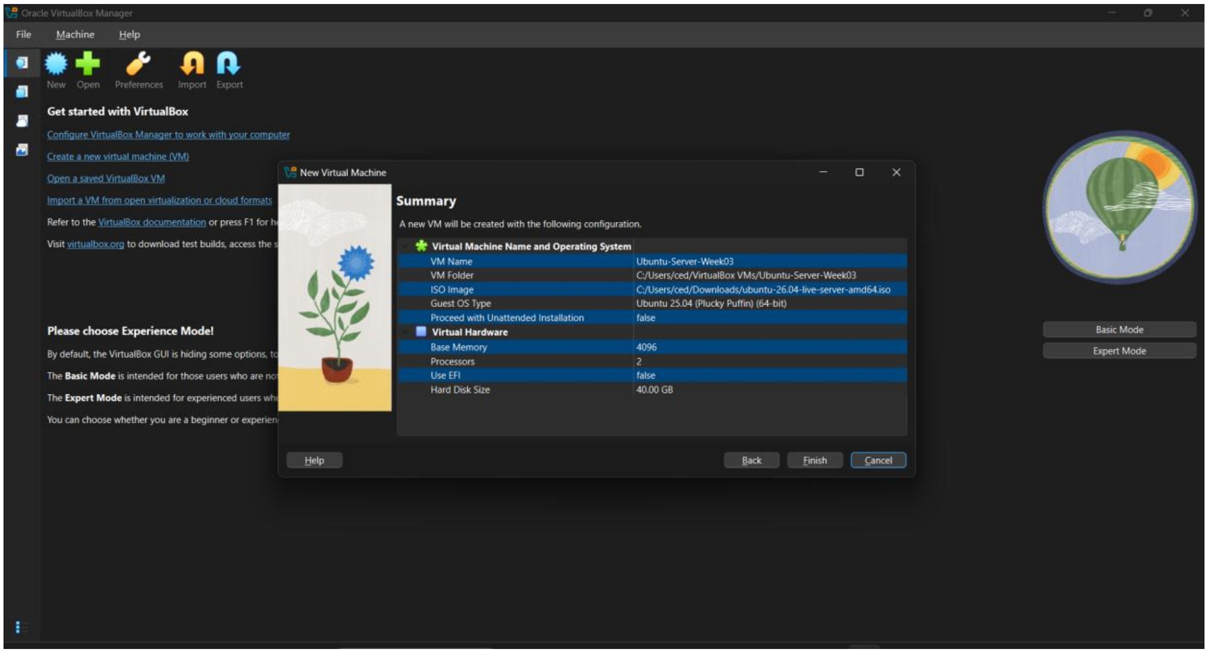
- Install and configure Ubuntu Server in a virtual machine.
- Configure hostname, keyboard, networking, storage, user account, and OpenSSH.
- Verify login, hostname, IP address, Internet access, system updates, and SSH.
- Compare BIOS and UEFI and document the Ubuntu boot process.
- Install and verify Windows Server Evaluation.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.

3. Software Requirements

- Oracle VirtualBox
- Ubuntu Server ISO: ubuntu-26.04-live-server-amd64.iso
- Windows Server 2025 Evaluation ISO

4. Virtual Machine Specifications

Component	Ubuntu Server Configuration
Name	Ubuntu-Server-Week03
RAM	4096 MB
CPU	2 virtual processors
Storage	40 GB dynamically allocated VDI
Network	NAT
Optical drive	Ubuntu Server ISO

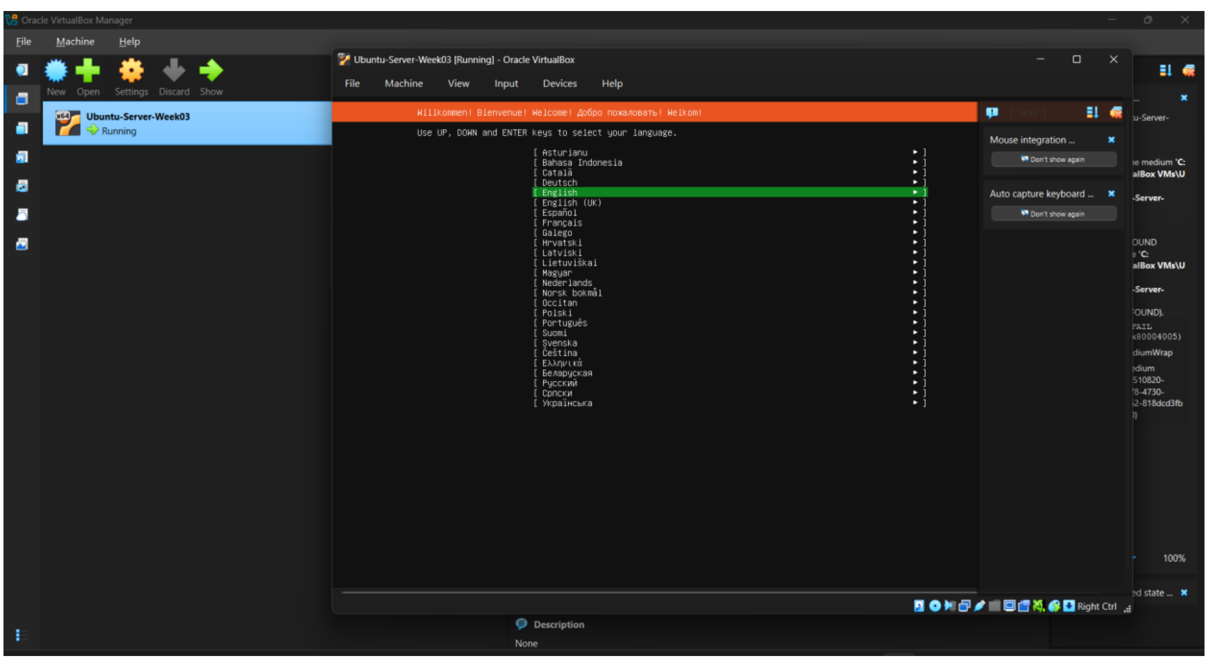


VirtualBox VM details showing the required hardware, Ubuntu Server ISO, 40 GB disk, and NAT network.

5. Ubuntu Server Installation Procedure

5.1 Language

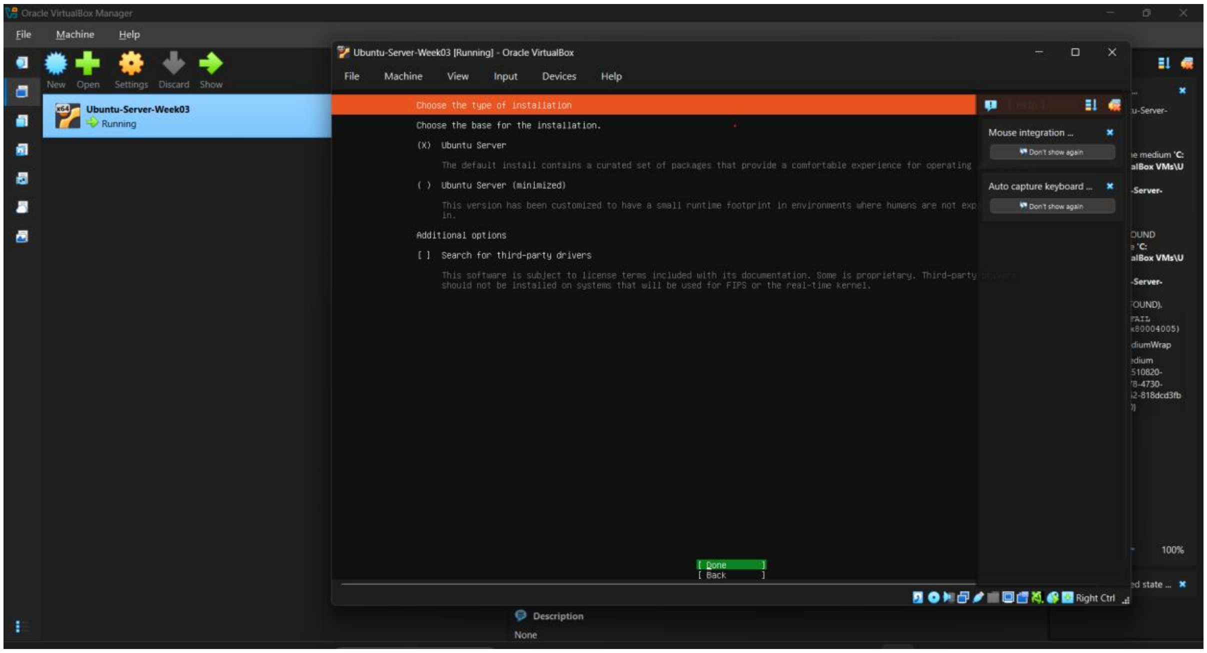
English selected.



Ubuntu Server language selection with English highlighted.

5.2 Installation type

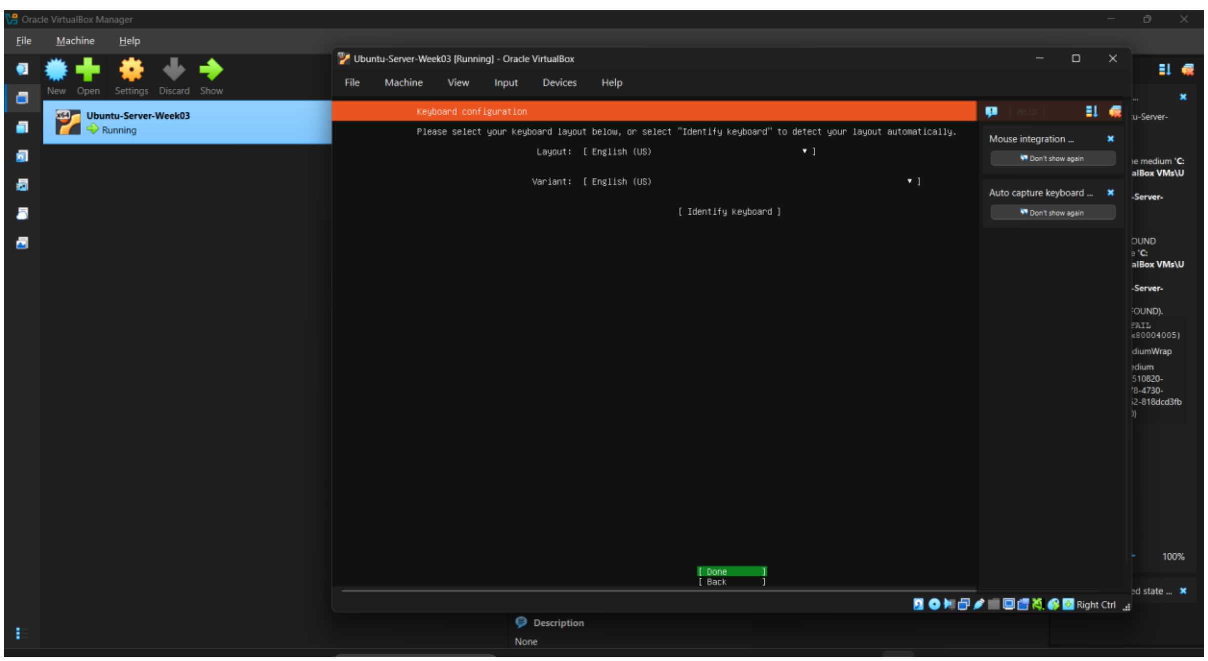
Standard Ubuntu Server selected; minimized installation and third-party drivers not selected.



Installation type configuration evidence.

5.3 Keyboard

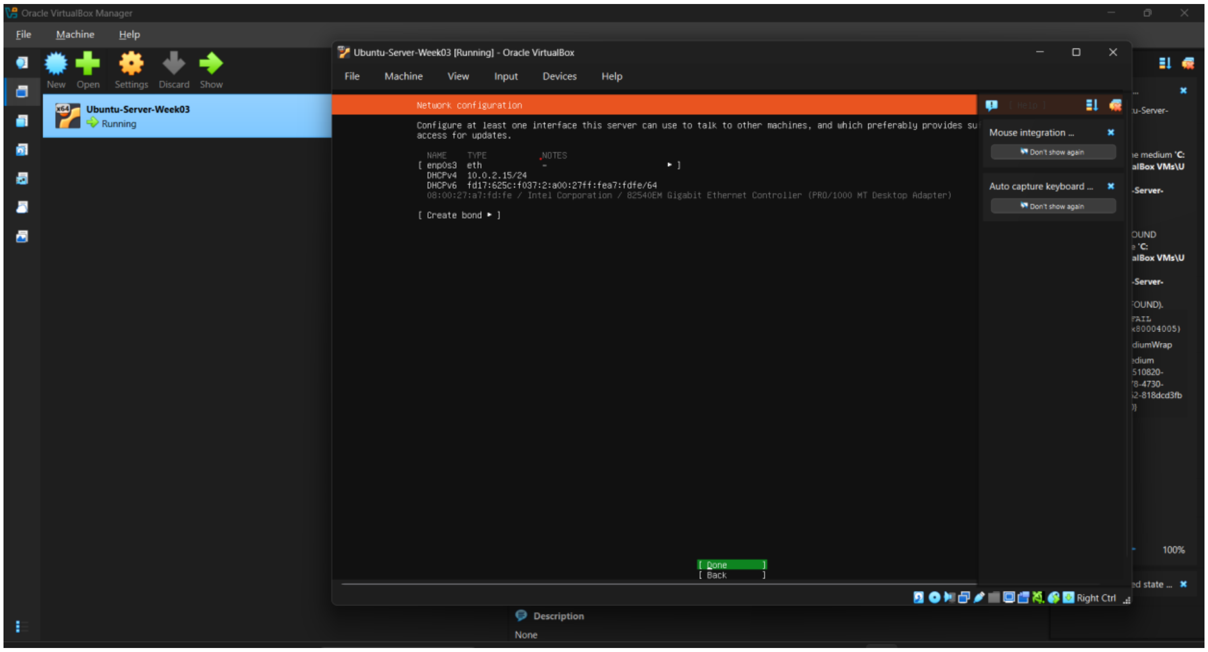
English (US) layout and variant selected.



Keyboard configuration evidence.

5.4 Network

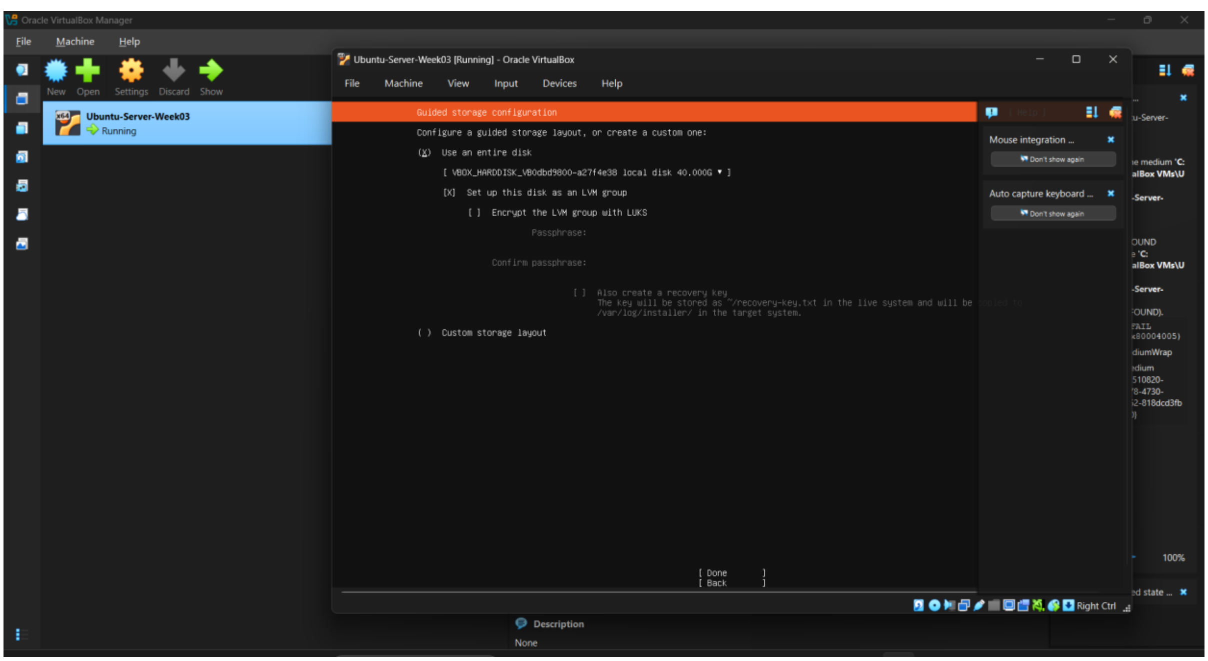
DHCP accepted; enp0s3 received 10.0.2.15/24.



Network configuration evidence.

5.5 Guided storage

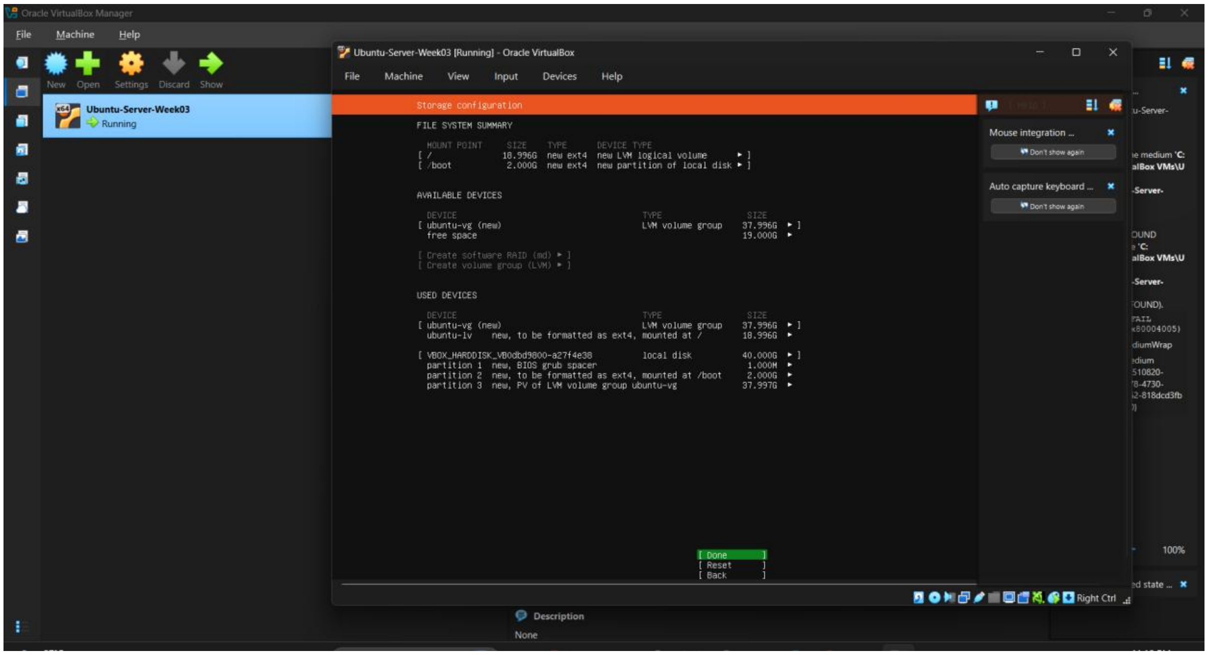
Use an entire disk selected with LVM.



Guided storage configuration evidence.

5.6 Storage summary

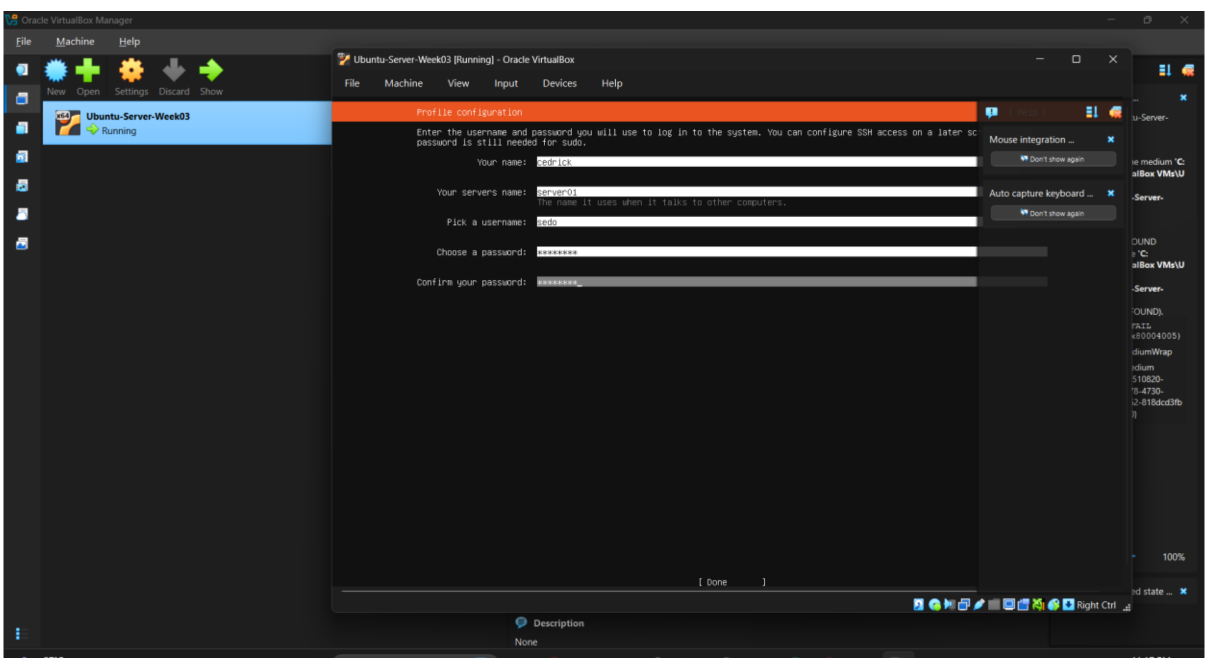
40 GB disk prepared with ext4 /boot and LVM root logical volume.



Storage summary configuration evidence.

5.7 Profile

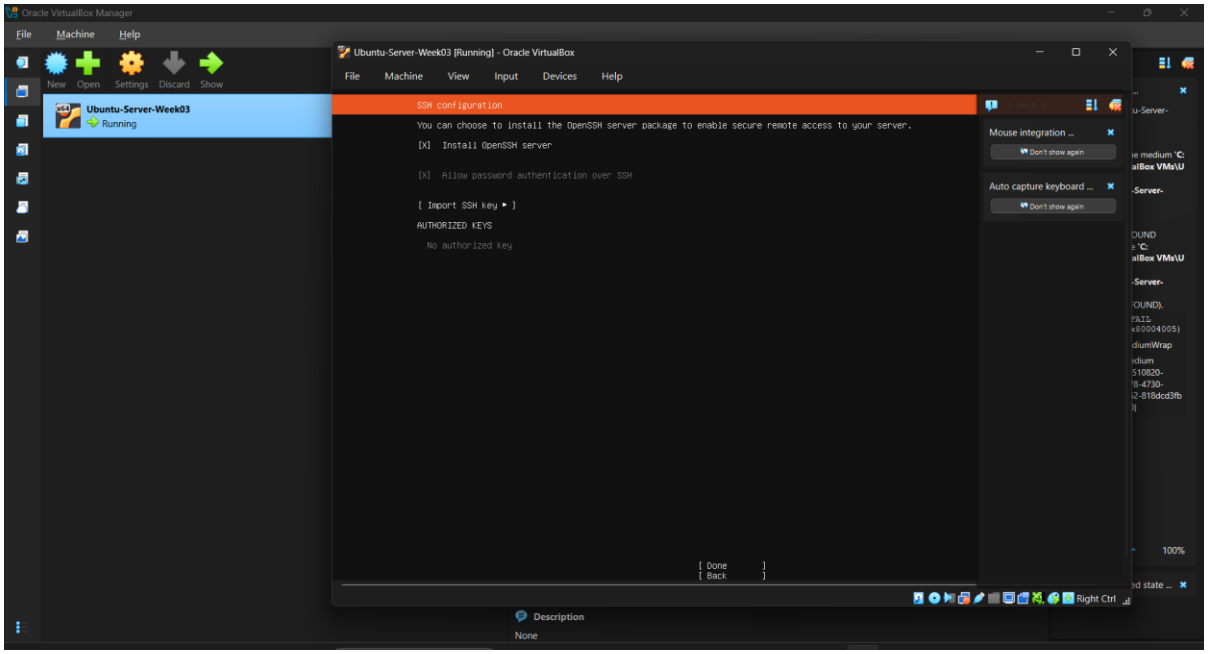
Hostname server01 and non-root administrative user sedo configured.



Profile configuration evidence.

5.8 SSH

Install OpenSSH server enabled.



SSH configuration evidence.

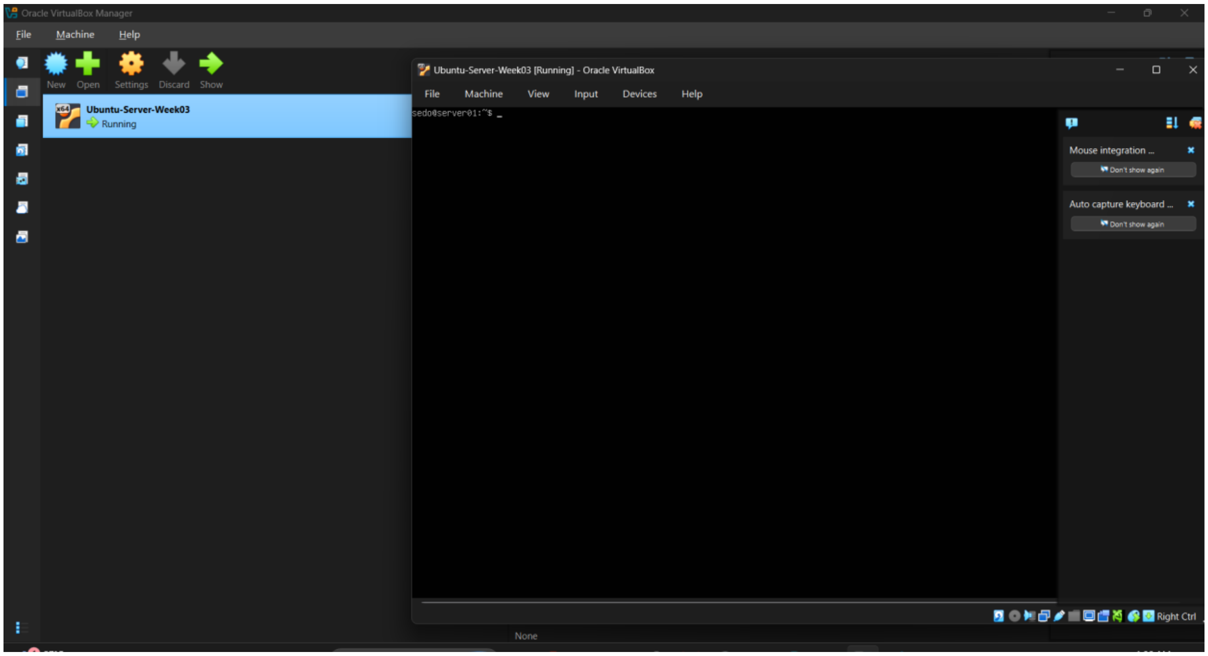
6. Configuration Details

Setting	Value
Hostname	server01
Administrative user	sedo
Network	NAT / DHCP
IPv4	10.0.2.15/24
Disk	40 GB, Guided whole disk with LVM
Filesystem	ext4 for / and /boot
SSH	OpenSSH Server installed
Additional packages	None selected

7. Server Verification

7.1 Login

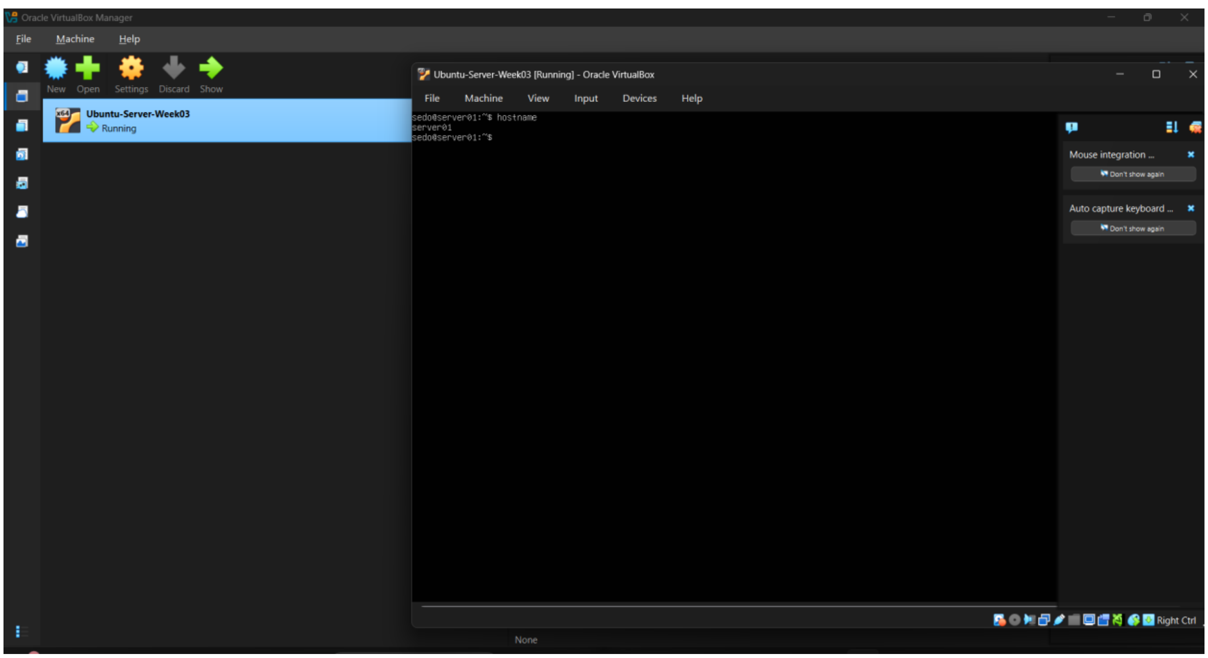
Successful as sedo.



Login verification evidence.

7.2 Hostname

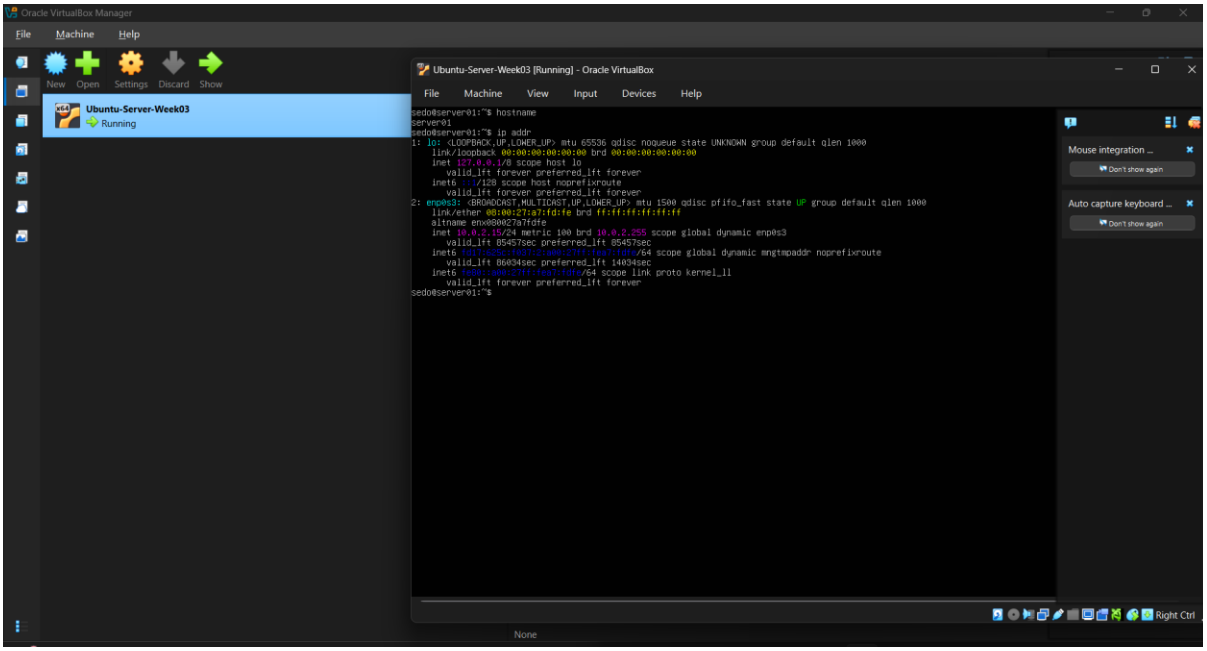
hostname returned server01.



Hostname verification evidence.

7.3 IP address

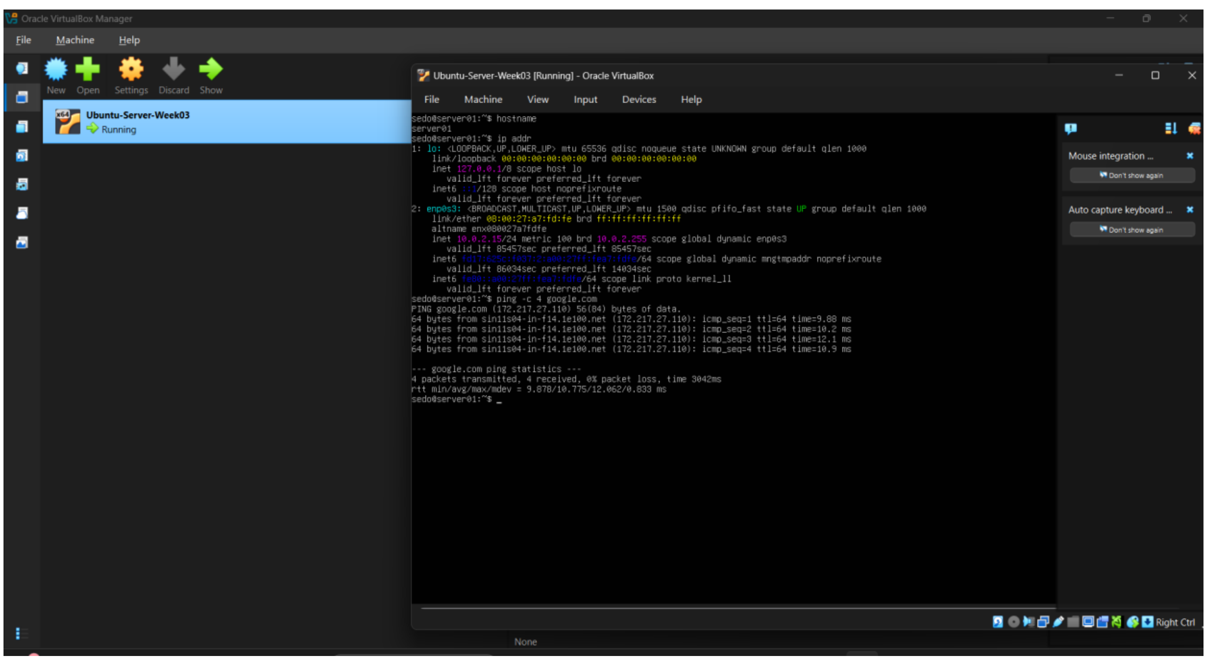
ip addr showed 10.0.2.15/24 on enp0s3.



IP address verification evidence.

7.4 Internet

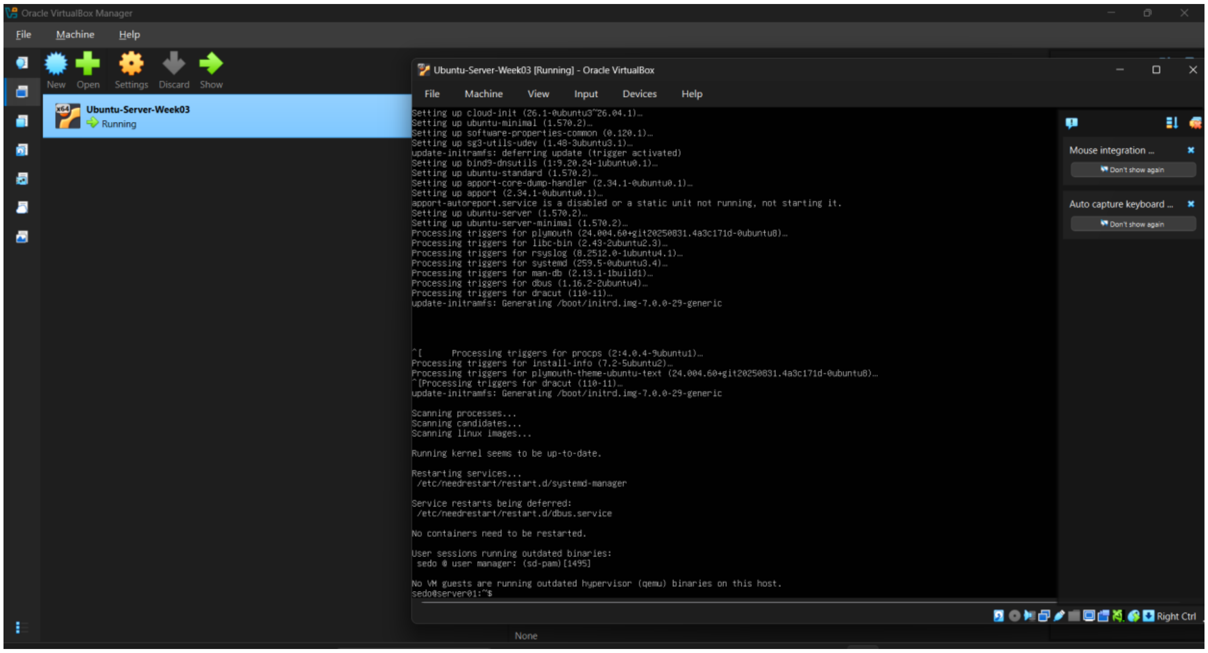
ping -c 4 google.com returned 4 replies and 0% packet loss.



Internet verification evidence.

7.5 Update

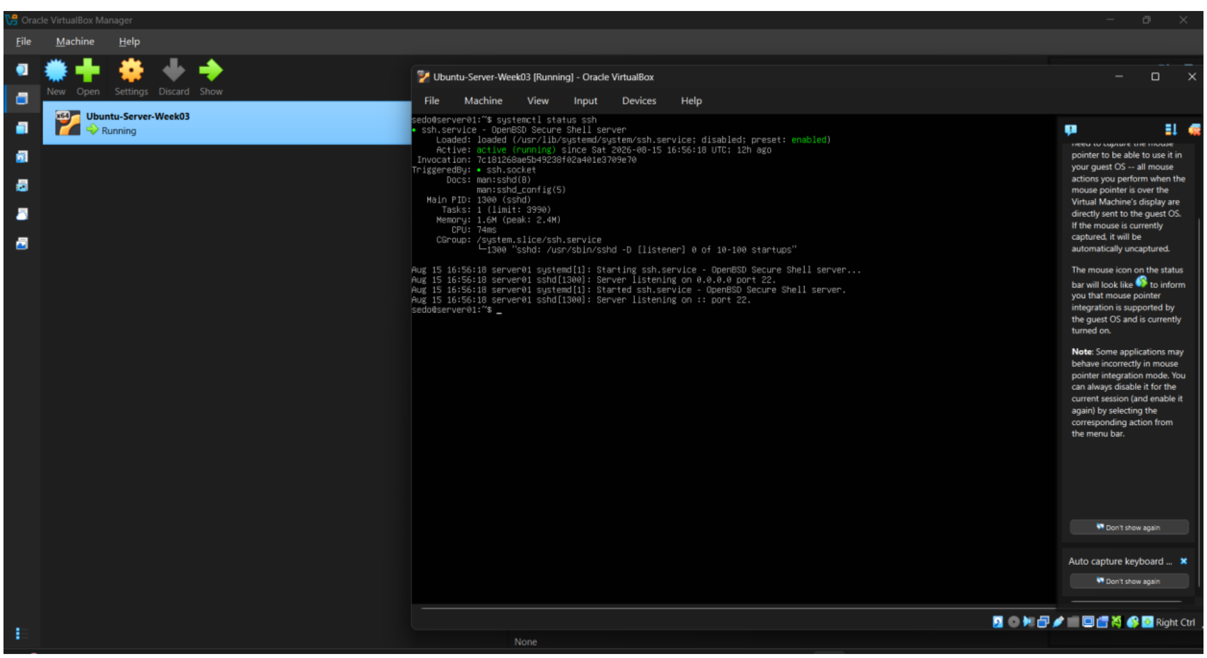
sudo apt update and sudo apt upgrade -y completed after repository troubleshooting.



Update verification evidence.

7.6 SSH

systemctl status ssh showed active (running), listening on port 22.



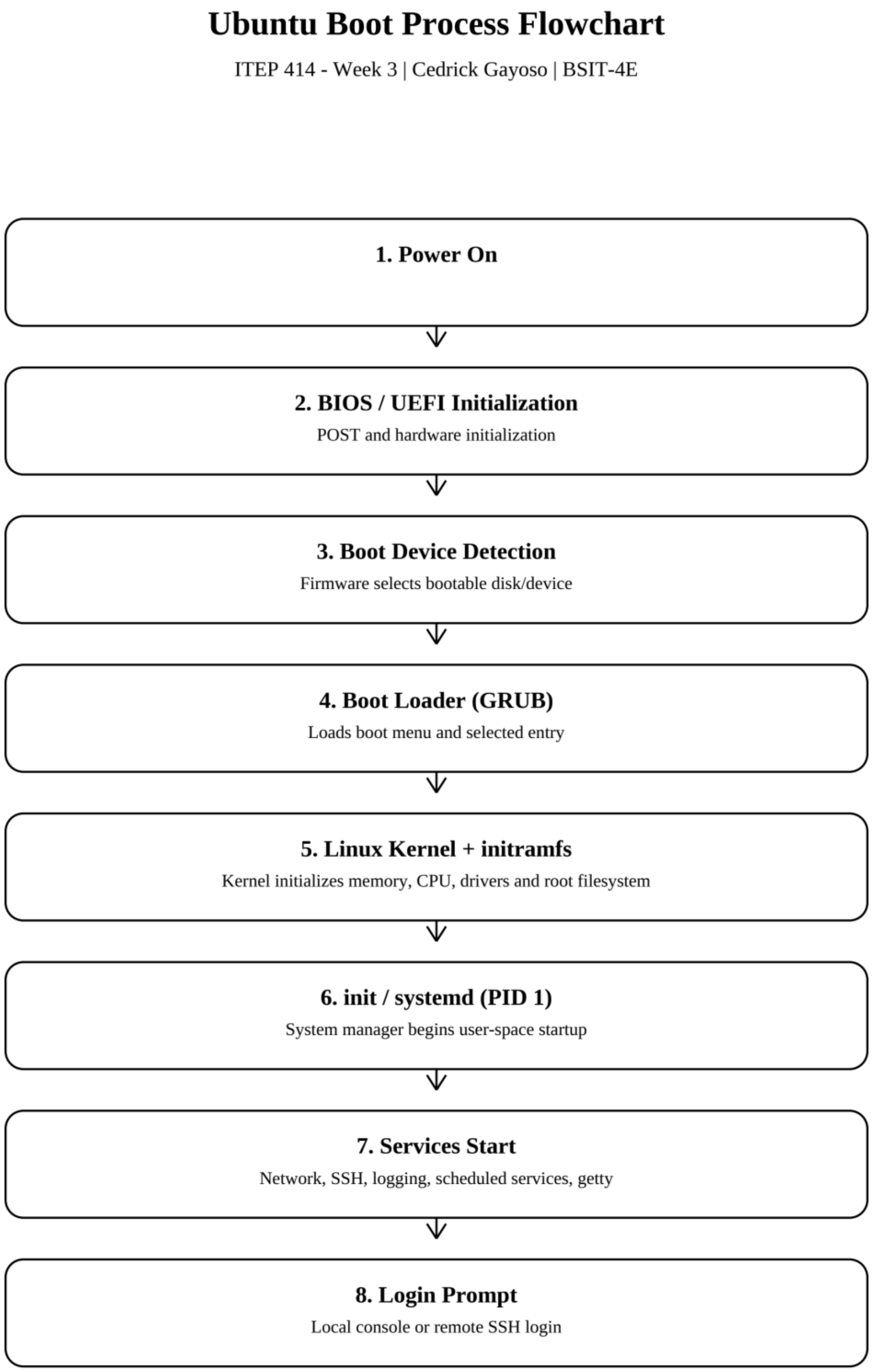
SSH verification evidence.

8. BIOS vs UEFI Comparison

Area	BIOS	UEFI
Definition	Legacy firmware interface.	Modern extensible firmware interface.
Boot process	Typically starts boot code from MBR/boot sector.	Loads EFI executable from EFI System Partition.
Maximum disk support	Legacy MBR commonly about 2 TiB limit.	Normally paired with GPT and supports much larger disks.
Partition style	MBR	GPT
Security	No standardized Secure Boot.	Supports Secure Boot.
Speed	Older initialization model.	Can provide more efficient boot management.
Advantages	Simple and compatible with older systems.	Modern disk support, flexible boot entries, stronger security options.
Disadvantages	Legacy limitations and fewer modern controls.	More complex firmware and Secure Boot troubleshooting.
Modern usage	Mostly legacy compatibility.	Standard on most modern PCs and servers.

UEFI has largely replaced BIOS because modern systems need larger-disk support, flexible boot management, and stronger firmware security. UEFI is normally paired with GPT and can launch EFI applications from the EFI System Partition. Secure Boot adds a firmware-level trust control before the operating system starts. BIOS remains useful for legacy compatibility and troubleshooting, but UEFI is generally preferred for current enterprise deployments.

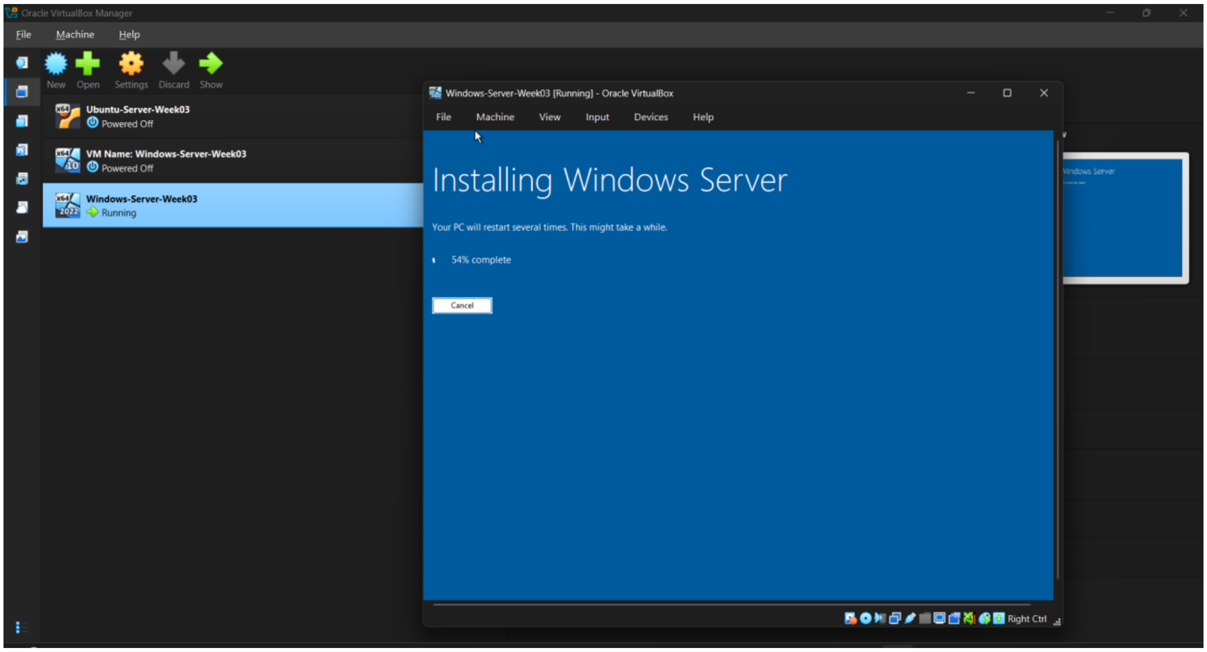
9. Ubuntu Boot Process Flowchart



Power On -> BIOS/UEFI -> Boot Device Detection -> GRUB -> Linux Kernel + initramfs -> init/systemd -> Services Start -> Login Prompt.

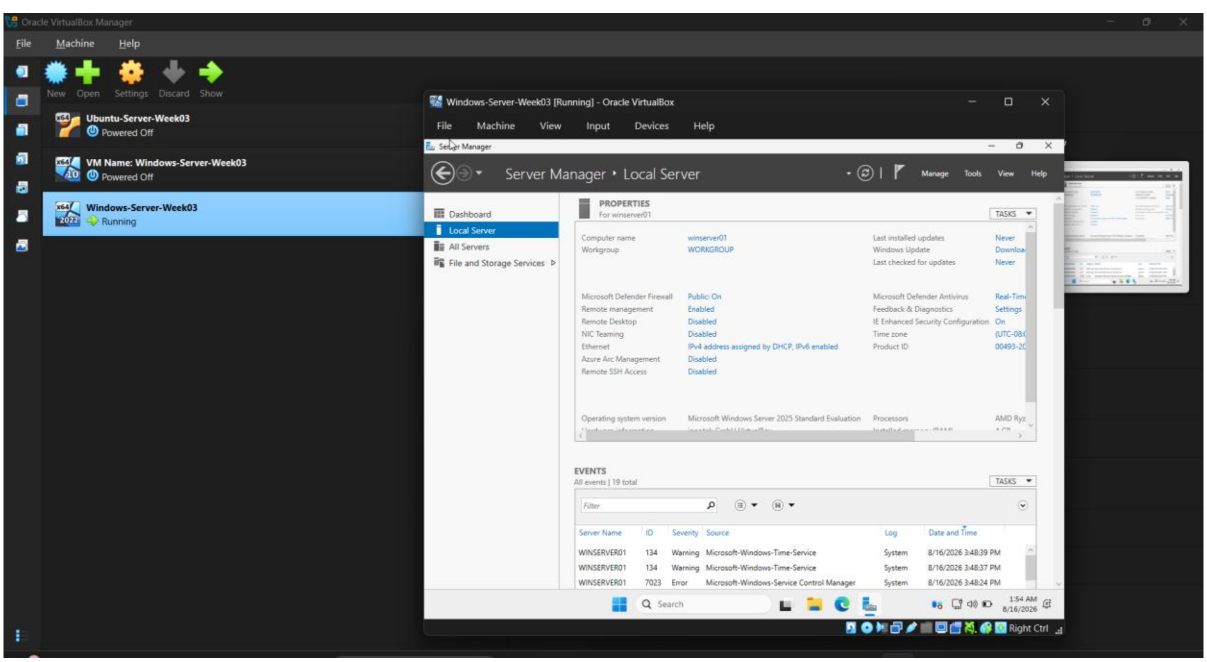
10. Windows Server Installation Summary

Windows Server 2025 Standard Evaluation (Desktop Experience) was installed in a separate VirtualBox VM with 4 GB RAM, 2 virtual CPUs, a 50 GB VDI, and NAT. During installation VirtualBox raised a BLKCACHE_IOERR while the host drive was low on free space. After freeing host storage and resetting the VM once, Windows Setup resumed successfully.



Windows Server installation evidence.

An Administrator password was created, the system was logged into successfully, and the computer name was changed to winserver01. Server Manager confirmed Windows Server 2025 Standard Evaluation and the final computer name.



Windows Server successful login and computer name winserver01.

11. Operating System Comparison

Area	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial/proprietary; evaluation available.	Open source; optional paid support.	Open source; community supported.
User interface	GUI or Server Core.	Primarily CLI.	Primarily CLI; GUI optional.
Package management	Windows Update, PowerShell, DISM.	APT/dpkg.	DNF/RPM.
Security	Defender, Firewall, Group Policy, AD controls.	AppArmor, OpenSSH, sudo, signed repositories.	SELinux, firewalld, OpenSSH, sudo.
Performance	Workload dependent; Server Core can reduce overhead.	Lightweight and popular for cloud/web/container workloads.	Stable RHEL-family enterprise platform.
Typical use	AD, IIS, Hyper-V, Microsoft workloads.	Web, cloud, DevOps, databases, containers.	RHEL-compatible enterprise infrastructure.
Advantages	Strong Microsoft integration.	Large ecosystem and broad cloud adoption.	Enterprise stability and SELinux integration.
Disadvantages	Commercial licensing and larger GUI footprint.	Linux CLI skills required.	Conservative packages and smaller general-purpose community.

The best platform depends on application compatibility, administrator skills, support, licensing, and ecosystem requirements. Windows Server fits Microsoft-centered infrastructure, Ubuntu Server is a flexible general-purpose Linux platform, and Rocky Linux is suitable for RHEL-family enterprise administration.

12. Problems Encountered and Solutions

Problem	Issue	Solution
Duplicate VDI	Disk/UUID already registered.	Stopped duplicate creation and retained the intended VM.
Missing VDI	VERR_FILE_NOT_FOUND.	Repaired/replaced invalid storage attachment.
Settings save issue	VM state/storage inconsistent.	Returned VM to powered-off state and repaired storage.

CPU soft lockup	Watchdog message during install.	Monitored and allowed installation to finish.
APT 403	Mirror could not provide a firmware package.	Changed to main archive, reran update and upgrade.
systemctl typo	systemct1 typed with digit 1.	Retyped systemctl status ssh correctly.
Windows I/O cache error	BLKCACHE_IOERR with low host free space.	Freed host storage, resumed/reset VM, installation continued.

13. Lessons Learned

A server deployment is complete only after identity, networking, storage, security, updates, verification, and documentation are confirmed. The project reinforced careful troubleshooting: read the exact error, preserve working data, change one variable at a time, and verify the result. It also showed how host resources such as disk space can affect a dynamically allocated virtual disk.

14. References

- Oracle VirtualBox User Manual.
- Ubuntu Server documentation.
- OpenSSH and systemd documentation.
- Microsoft Windows Server documentation.
- UEFI Forum documentation.
- Rocky Linux documentation.